

		Hence $d = 40$
24	B	e.m.f. of cell, $E = 1.5 \text{ V}$ (From Fig. a) Internal resistance of cell, $r = 1.5 / 5.0 = 0.3 \text{ } \Omega$ (From Fig. b) Current in Fig. c, $I = 1.5 / (1.2 + 0.3) = 1.0 \text{ A}$ Voltmeter reading for Fig. c $= (1.0)(1.2) = E - I r = 1.5 - (1.0)(0.3) = 1.2 \text{ V}$
25	C	Since the charges at X and Y are equal and opposite, the magnitude of the electric field